

Chapter 3. **GROUPS**

3.1 What is a group?

3.1.2 Inverses. Exponent Laws. Cancellation Laws



Today we will

- Define the powers of an element g of a group G ;
- Prove some important properties, known as the
 - **EXPONENT LAWS**
 - **CANCELLATION LAWS**

How can we define the powers of an element?

Let G be a group and $g \in G$. If $n \geq 0$ denotes an integer, we can inductively define the **n th power** g^n of g as follows:

$$g^0 = 1, \quad g^n = g \cdot g^{n-1}$$

The definition of n th powers could be extended to include **negative powers** by considering the inverse of g . More precisely, we define

$$g^{-n} = (g^{-1})^n = (g^n)^{-1}$$

We can now talk about the **ORDER** of an element:

Definition 3.6. Let G be a group and $g \in G$. The **order** of the element g is defined as the **smallest positive integer** n such that $g^n = 1$. If $g^k \neq 1$ for all $k \geq 1$, then we say that g has **infinite order**. We write $\text{o}(g)$ (or sometimes $|g|$) to denote the order of g .

Examples 3.7. ^① All the elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$ have order 2.

$$2 \cdot (\bar{1}, \bar{0}) \stackrel{?}{=} (\bar{0}, \bar{0})$$

$$(\bar{1}, \bar{0}) + (\bar{1}, \bar{0}) = (\bar{1} + \bar{1}, \bar{0} + \bar{0}) = (\bar{1} + \bar{1}, \bar{0} + \bar{0}) = (\bar{0}, \bar{0})$$

$$\textcircled{2} \quad \mathbb{Z}_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$$

$$2 \cdot \bar{3} = \bar{3} + \bar{3} = \bar{6} = \bar{2} \quad ; \quad 3 \cdot \bar{3} = \bar{3} + 2 \cdot \bar{3} = \bar{3} + \bar{2} = \bar{5} = \bar{1}$$

$$4 \cdot \bar{3} = \bar{3} + 3 \cdot \bar{3} = \bar{3} + \bar{1} = \bar{4} = \bar{0}$$

③ $1 \in \mathbb{Z}$ has infinite order

④ $o((1\ 2)) = 2$ in S_3

Inverses. Exponent Laws. Cancellation Laws

THEOREM 3.8. Let $g, h, g_1, \dots, g_n$ denote elements of a group G .

- (1) $1^{-1} = 1$.
- (2) $(g^{-1})^{-1} = g$.
- (3) $(gh)^{-1} = h^{-1}g^{-1}$. (G might not be abelian)
- (4) $(g_1g_2 \dots g_{n-1}g_n)^{-1} = g_n^{-1}g_{n-1}^{-1} \dots g_2^{-1}g_1^{-1}$ for all $n \geq 1$.

THEOREM 3.9. Exponent Laws. Let G be a group and $g, h \in G$.

- (1) $g^n g^m = g^{n+m}$ for all $n, m \in \mathbb{Z}$.
- (2) $(g^n)^m = g^{nm}$ for all $n, m \in \mathbb{Z}$.
- (3) If $gh = hg$, then $(gh)^n = g^n h^n$ for all $n \in \mathbb{Z}$.
 g, h commute

THEOREM 3.10. Let g, h , and f be elements of a group.

- (1) If $gh = gf$, then $h = f$. **Left cancellation Law**
- (2) If $hg = fg$, then $h = f$. **Right cancellation Law**
- (3) The equation $gx = h$ has a unique solution $x = g^{-1}h$ in G .
- (4) The equation $xg = h$ has a unique solution $x = hg^{-1}$ in G .

. Proof of (3) in Thm 3.9

Exercise. Suppose that $a, b \in G$ such that $ab = ba$. Use induction to show that

① $a^n b^n = b^n a^n$ for all $n \geq 0$ $\leftarrow$

② $ab^n = b^n a$ for all $n \geq 0$

Suppose that $gh = hg$.

Goal. $(gh)^n = g^n h^n$ for all $n \in \mathbb{Z}$.

$n=0$ Trivially
works.

We first prove it for $n > 0$.

$n=1$. $(gh)^1 = gh$ ✓

Assume that $(gh)^{n-1} = g^{n-1} h^{n-1}$.

$$\begin{aligned} (gh)^n &= (gh)(gh)^{n-1} \stackrel{\text{I.H.}}{=} (gh)(g^{n-1} h^{n-1}) \\ &= \underline{g(hg^{n-1})} h^{n-1} = \underline{g(g^{n-1}h)} h^{n-1} \\ &= (gg^{n-1})(hh^{n-1}) = \underline{g^n h^n} \end{aligned}$$

If $n < 0$, then $-n > 0$ and we know

that $(gh)^{-n} = \underline{g^{-n} h^{-n}} \stackrel{\text{Exercise}}{=} \underline{h^{-n} g^{-n}} \stackrel{\text{Exercise}}{=} \underline{g^{-n} h^{-n}}$

Thus:

$$\left((gh)^{-n} \right)^{-1} = \left(g^{-n} h^{-n} \right)^{-1} \stackrel{\text{Exercise}}{=} \left(h^{-n} g^{-n} \right)^{-1}$$

$$\textcircled{1} \underline{\left((gh)^{-n} \right)^{-1}} \stackrel{\text{def.}}{=} \underline{\left(\left((gh)^{-n} \right)^{-1} \right)^{-1}} = \underline{(gh)^n}$$

$$\begin{aligned} \left(h^{-n} g^{-n} \right)^{-1} &= \left((h^n)^{-1} (g^n)^{-1} \right)^{-1} = \\ &= \left((g^n)^{-1} \right)^{-1} \left((h^n)^{-1} \right)^{-1} = \underline{g^n h^n} \end{aligned}$$

We have proved that

$$\underline{(gh)^n} = \left((gh)^{-n} \right)^{-1} = \left(h^{-n} g^{-n} \right)^{-1} = \underline{g^n h^n}$$

Exponent Laws in additive notation:

Multiplicative notation

$$g^{n+m} = g^n g^m$$

$$(g^n)^m = g^{nm}$$

$$(gh)^n = g^n h^n$$

if g and h commute

Additive notation

$$(n+m)g = ng + mg$$

$$m(ng) = (mn)g$$

$$n(g+h) = ng + nh$$

A very useful property of the Cayley table:

COROLLARY 3.11. *Every row (and column) of the Cayley table of a finite group G contains every element of G exactly once.*

Proof. For $g \in G$, the row of the Cayley table corresponding to g consists of the elements of the form gx , where $x \in G$. Such row contains $h \in G$ exactly once because the equation $gx = h$ has a unique solution in G . $\square$